



1. (a) Write the SQL query to create the Department table with the following constraints.

Department (deptId, deptName, location, deptHead, numEmployees)

deptId is a number used as the primary key

deptName cannot be null

Default location to be 'Galle'

deptHead specifies a unique number

numEmployees should be a number between 1-50

```
CREATE TABLE Department (  
    deptId int Primary Key,  
    deptName CHAR(20) NOT NULL,  
    location CHAR(25) DEFAULT 'Galle',  
    deptHead INT UNIQUE,  
    numEmployees INT CHECK (numEmployees BETWEEN 1 AND 50));
```

(b) Consider the following relational schema to answer the given question.

Employee (eno, ename, hiredate, salary, saldifference)

Write a trigger to calculate and update the saldifference filed (difference between the old and new salary) when the salary is updated.

```

DELIMITER //
CREATE TRIGGER SalUpdate
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    DECLARE sal_dif INT;

    IF NEW.salary > OLD.salary THEN
        SET sal_dif = NEW.salary - OLD.salary;
    ELSE
        SET sal_dif = OLD.salary - NEW.salary;
    END IF;

    UPDATE Employee SET saldifference = sal_dif WHERE eno =
NEW.eno;
END//

```

(c) Write down two (02) common uses of database triggers.

- Provide sophisticated auditing
 - Prevent invalid transactions
 - Enforce referential integrity (either those actions not supported by declarative constraints or across nodes in a distributed database)
 - Enforce complex business rules
 - Enforce complex security authorizations
 - Provide transparent event logging
 - Automatically generate derived column values
 - Enable building complex views that are updatable
 - Track database events
- Any two of the above.

(d) Suppose there is a relation declared as follows:

```

CREATE TABLE Employee (Name VARCHAR(50) PRIMARY KEY,
Salary INT CHECK(Salary <= 40000) );

```

Initially, the relation has three records:

Name	Salary
Perera	10000
Costa	20000
Silva	30000

We execute the following sequence of modifications.

(i) INSERT INTO Employee VALUES ('Dias', 12000);

(ii) UPDATE Employee SET Salary = 50000 WHERE Name = 'Silva';

(iii) INSERT INTO Employee VALUES ('Perera', 13000);

(iv) DELETE FROM Employee WHERE Name = 'Costa';

At the end of these statements, calculate the sum of the Salaries over all the tuples in the Employee relation.

52000

(e) Consider the tables Employee and Evaluation as given below to keep track of employees, and their job evaluation records.

Employee (Empid, First_name, Job_id, Salary, Department_id)

Evaluation (Empid, Evaluation_date , Job_id, Department_id, Score)

Write down a trigger named **Salary Increment** to increase salary of an employee as given below (based on the Score value) upon insertion of his/her evaluation record into the Evaluation table.

20% salary increment if score is ≥ 8

10% salary increment if score ≥ 6 and score < 8

```

DELIMITER $$

CREATE TRIGGER Salary_Increment
AFTER INSERT ON Evaluation
FOR EACH ROW
BEGIN
    IF (NEW.Score >= 8) THEN
        UPDATE Employee
        SET Salary = Salary * 1.20
        WHERE Empid = NEW.Empid;
    ELSEIF (NEW.Score >= 6 AND NEW.Score < 8) THEN
        UPDATE Employee
        SET Salary = Salary * 1.10
        WHERE Empid = NEW.Empid;
    END IF;
END$$

DELIMITER ;

```

2) (a)

Suppose that we have an ordered file with $r = 30,000$ records stored in a disk with block size $B = 1024$ bytes. File records are of fixed size and unspanned, with record size $R = 100$ bytes.

(i) Calculate the blocking factor (BFR).

$$\begin{aligned}
 \text{BFR} &= \lfloor 1024/100 \rfloor \\
 &= 10 \text{ records per block.}
 \end{aligned}$$

(ii) How many block accesses are required to search a record in the data file using the binary Search?

$$\begin{aligned}
 \text{The number of blocks needed for the file is } b &= \lceil (r/\text{bfr}) \rceil \\
 &= \lceil (30,000/10) \rceil = 3000 \text{ blocks.}
 \end{aligned}$$

$$\begin{aligned}
 \text{A binary search on the data file would need approximately } \lceil \log_2 b \rceil \\
 = \lceil (\log_2 3000) \rceil = 12 \text{ block accesses.}
 \end{aligned}$$

(iii) Suppose that the ordering key field of the file is $V = 9$ bytes long, a block pointer is $P = 6$ bytes long, and we have constructed a primary index for the file. How many block accesses are required to search a record using the primary index?

Size of index entry is $9+6 = 15$ bytes.

Blocking factor for index = $L(1024/15) = 68$

Number of index entries is 3000

The number of index blocks is $= \lceil (3000/68) \rceil = 45$

To perform a binary search on index file would need $\lceil (\log_2 b) \rceil = \lceil (\log_2 45) \rceil = 6$ block accesses.

Total number of block access = $6 + 1$ block accesses = 7 block accesses

(b) Consider the following relational schema.

Employee (EmpId, Name, Address, Gender, Marital_Status, Dept_Id, Salary)

What is the most suitable index structure to efficiently retrieve the details of employees based on their Marital_Status and Gender?

Bitmap index

(c) Write down the difference between the **Dense Index** and the **Sparse Index**.

Dense Index - Dense index contains an index record for every search key value in the data file.

Sparse Index - Instead of pointing to each record in the main table, the index points to the records in the main table in a gap (index item points to a block).

(d) Briefly explain the index only strategy technique using a suitable example.

A technique used by databases to retrieve data using only the index, without having to access the actual table data.

Suitable Example

(e) Concurrent executions of transactions can cause problems for the database. Briefly explain the **Lost Update problem** and **Temporary Update problem** that can occur due to the concurrent execution of transactions. Use appropriate examples to describe the two problems.

Temporary Update (Dirty Read) problem

Occurs when one transaction updates a database item and then the transaction fails for some reason. Meanwhile.

T1	T2
READ(X) $X = X - N$ WRITE(X)	
	READ (X) $X = X + M$ WRITE(X)
READ(Y) ROLLBACK	

Lost Update problem

Occurs when two different transactions are trying to update the same column on the same row within a database at the same time.

T1	T2
READ(X) $X = X - N$	
	READ (X) $X = X + M$
WRITE(X) READ(Y)	
	WRITE(X)

3. (a) State two (02) advantages of distributed databases.

- Reflects organizational structure
- Local autonomy
- Improved availability
- Improved performance
- Modularity

(b) State two (02) disadvantages of distributed databases.

- Complexity
- Economics
- Security
- Difficult to maintain the integrity
- Inexperience

(c) Draw the architectural diagram for a distributed database management system and identify its components.

